

Musculoskeletal and Emergency Imaging

Desmoid-type fibromatosis: Tumour response assessment using magnetic resonance imaging signal and size criteria

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ABSTRACT

Introduction: This study aimed to establish a specified magnetic resonance imaging (MRI) signal and size criterion for assessing the response of desmoid-type fibromatosis (DF).

Methods: This retrospective study included 129 patients with DF who received non-surgical therapy. All patients underwent pretreatment and 6-month-interval follow-up MRI for > 3 years (6 follow-up visits). The correlation between signal grade and size was determined. Signal grade and size among three response groups (partial response [PR], stable disease [SD], progression disease [PD]) were compared. The specified signal and size criterion was established, used to assess tumour response at each follow-up, and compared with the reference. The Response Evaluation Criteria in Solid Tumours (RECIST)1.1 criterion at the end of the 3rd year was considered the reference.

Results: MRI signals were moderately correlated with size changes ($r = -0.56$ and -0.41 for T2 grade and contrast-enhanced T1 grade, respectively). Changes in T2 grade and size in the three response groups were significantly different (all $p < 0.01$). The signal and size criterion accurately predicted 95% of PR patients at 2nd follow-up and 81.2% of PD patients at the 3rd follow-up, while only 13.1% of PR and 56.3% of PD patients were predicted by RECIST1.1. However, the accuracy of the signal & size criterion for predicting SD was lower than that of RECIST1.1.

Conclusions: MRI signal is useful in assessing the response of DF. Signal & size criterion can identify patients with PR and PD earlier than RECIST1.1.

1. Introduction

Desmoid-type fibromatosis (DF) is a locally aggressive, nonmetastatic mesenchymal tumour that is often difficult to resect. Although these intermediate-grade tumours lack the potential to metastasize, they may cause serious morbidity because of their infiltrative growth and severe tendency to local recurrence [1–4]. Immediate surgery is no longer considered the standard treatment of DF [1,5,6]. There are various non-surgical management methods, including hormonal agent, anti-inflammatory drug, cytotoxic agent, and molecular targeted therapies, radiation therapy, and radiofrequency ablation [1,5]. It remains a challenge for clinicians to decide the best therapy.

The size-based Response Evaluation Criteria in Solid Tumours (RECIST) and WHO criteria are often employed in clinical practice [5,7], but there are some recognised limitations [8–11]. Several studies

have indicated that size-based response evaluation criteria are insensitive to the biological response observed in some soft-tissue sarcomas [10,12]. DF has a unique histology but becomes biologically quiescent, which is reflected by a decrease in T2 signal intensity and enhancement [1,5,13,14]. Some studies have shown inherent limitations in the RECIST1.1 evaluation of DF, which could have an impact on clinical management [13,14]. However, there is no recognised criterion for identifying good responders to DF [13–15]. Consequently, a well-tailored set of response evaluation criteria, not only considering size change, needs to be constructed in response assessments for DF.

MRI is widely used in the diagnosis and follow-up of DF [1,5,6]. Some studies have shown that MRI signal is a useful tool in clinical decision-making in patients with DF [16]. A criterion proposed by Braschi-Amirfarzan et al. [1] uses both size and MRI signal intensity to assess the treatment response of DF. However, this criterion does not

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quantify the changes in size and MRI signals of tumours. Recent studies [17,18] indicated that RECIST may underestimate the efficacy of DF treatment, and a better criterion may be MRI signal intensity. In this study, we adapted the criterion proposed by Braschi-Amirfarzan et al. into clinical practice according to previous studies [10,15,19].

This study aimed to analyse changes in MRI signals in the tumour response assessment of DFs, establish a specific combination of MRI signal and size criteria, and evaluate the accuracy and application of this combined criterion.

2. Materials and methods

This retrospective, non-interventional study was approved by the institutional review board of our hospital. The requirement for informed consent was waived.

2.1. Patients

The study included consecutive patients from the database of patients with DF in our hospital from September 2009 to December 2016. The inclusion criteria were (a) solitary lesion (a lesion was considered multifocal if at least 1 cm of definitively normal tissue on all available sequences separated the different components) [20]; (b) close follow-up or non-surgical therapy; (c) pre-treatment MRI and at least a 6-month-interval MRI follow-up with similar protocols for at least 3 years [2,4]. All patients underwent at least seven MR examinations at baseline and 6th, 12th, 18th, 24th, 30th, and 36th month of follow-up. If a patient underwent more MR examinations, the latest one to the arranged time was adopted. The exclusion criteria were (a) follow-up MR examinations for < 3 years, one or more missed follow-up MRI, or missed MR examination at the arranged time of follow-up (± 1 month was acceptable); (b) multiple lesions; (c) surgery during follow-up; (d) insufficient axial T1 weighted imaging (WI)/T2WI, coronal T2WI, or axial contrast-enhanced (CE) T1WI sequences.

Finally, 129 patients were enrolled (Appendix 1). Patients' clinical information, including sex and age at presentation, and tumour type (primary or recurrence) were obtained from the hospital electronic patient record system.

2.2. Management strategies

There are four management options: (a) wait-and-see policy with close imaging follow-up every 3–6 months [2,4] until progressive disease (PD) was observed; (b) cytotoxic therapy, vinorelbine and methotrexate; (c) molecular targeted therapies, imatinib; d. mixed strategies, cytotoxic and molecular targeted therapies. The non-surgical therapy plans were decided by the Sarcoma Multidisciplinary Board for each patient.

2.3. MRI protocol

MRI was performed in 129 patients using a 1.5-T MRI scanner (Siemens Magnetom Aera, Germany) with dedicated coils. The imaging protocol contained the following sequences: axial fat-suppressed T1WI, axial T2WI, axial fat-suppressed T2WI, axial fat-suppressed T1 CE imaging, and coronal or sagittal fat-suppressed T1 CE imaging, and coronal or sagittal T2WI. The axial fat-suppressed T2WI parameters were as follows: repetition time (TR), 5000–5560 ms; echo time (TE), 74–85 ms; averages, 1–2; bandwidth, 279–422 Hz/pixel; turbo factor, 14–16; slice thickness, 5–7 mm; gap, 1–1.4 mm; matrix, 320×256 – 320×288 ; and field of view (FOV), 200–360 mm. The axial fat-suppressed T1 CE imaging parameters were as follows: TR, 6.7–6.84 ms; TE, 2.39 ms; flip angle, 10; averages, 2; bandwidth, 600 Hz/pixel; slice thickness, 4–5 mm; matrix, 288×259 – 320×288 ; and FOV, 260–360 mm. Acquisition protocols were adapted to patient anatomy and tumour location.

2.4. Radiological evaluation of tumour response using size and signal(S & S) criteria

The baseline and follow-up MR images from all patients were retrospectively evaluated by two radiologists (ZHC and JWY with 8 and 3 years' experience in interpreting MR images). The observers were informed of the baseline MR images which were labelled, but were blinded to the treatment regimen and time of follow-up MRI. MR images were evaluated on a picture archiving and communication system (PACS) workstation (Carestream Health v.12.0; Rochester, NY). The location of the tumour and involvement of adjacent structures, including the neurovascular bundle and bone, were recorded by the two radiologists working together [21]. If there was a discrepancy, a third experienced radiologist (SYS, with 25 years' experience in interpreting MR images) was included for arbitration. The two radiologists (ZHC and JWY) independently measured the tumour diameter and signal intensity. Tumour therapeutic response assessment was conducted according to RECIST 1.1 [22] and signal & size criterion (S & S) [1] at each time point, respectively (Table 1).

2.4.1. RECIST 1.1

The measurements of greatest diameters were conducted in any plane on T2WI or CE T1WI. Peritumoral oedema was not considered part of the tumour diameter. The variations in diameter of the two radiologists were averaged. Tumours were then divided into four groups: complete response (CR), partial response (PR), stable disease (SD), or progressive disease (PD).

2.4.2. S & S

To evaluate signal intensity, the radiologists defined the whole tumour areas and areas of signal hyperintensity by manually drawing on each slice of stacked MR images on axial T2 with fat suppression and

Table 1
Tumour response according to RECIST1.1 and signal and size criterion.

Response	RECIST1.1	Signal and size criterion(S&S) ^a
Complete response (CR)	Disappearance of all lesions	Disappearance of all lesions
Partial response (PR)	$\geq 30\%$ decrease in the greatest diameters	Decrease in size of the tumour of $> X\%$ with or without change in signal intensity at MR imaging; or unchanged/increased tumour size of $> X\%$ but decreased T2 signal intensity and enhancement at MR imaging.
Stable disease (SD)	Does not meet criteria for complete response, partial response, or progressive disease	Unchanged size (the change in size was $< X\%$) and signal intensity characteristics
Progressive disease (PD)	$\geq 20\%$ increase in the greatest diameters; new lesions	Increase in size of $> X\%$ without signal intensity changes in MR imaging; or unchanged tumour size but increased T2 signal intensity and enhancement at MR imaging.

^a According to different size variations, the signal and size criterion were labelled S & S 5%, S & S 10%, S & S 15%, S & S 20%, S & R, respectively. S & R referred to size variation of 30% in the PR group and 20% in the PD group. S & S = Signal and size.

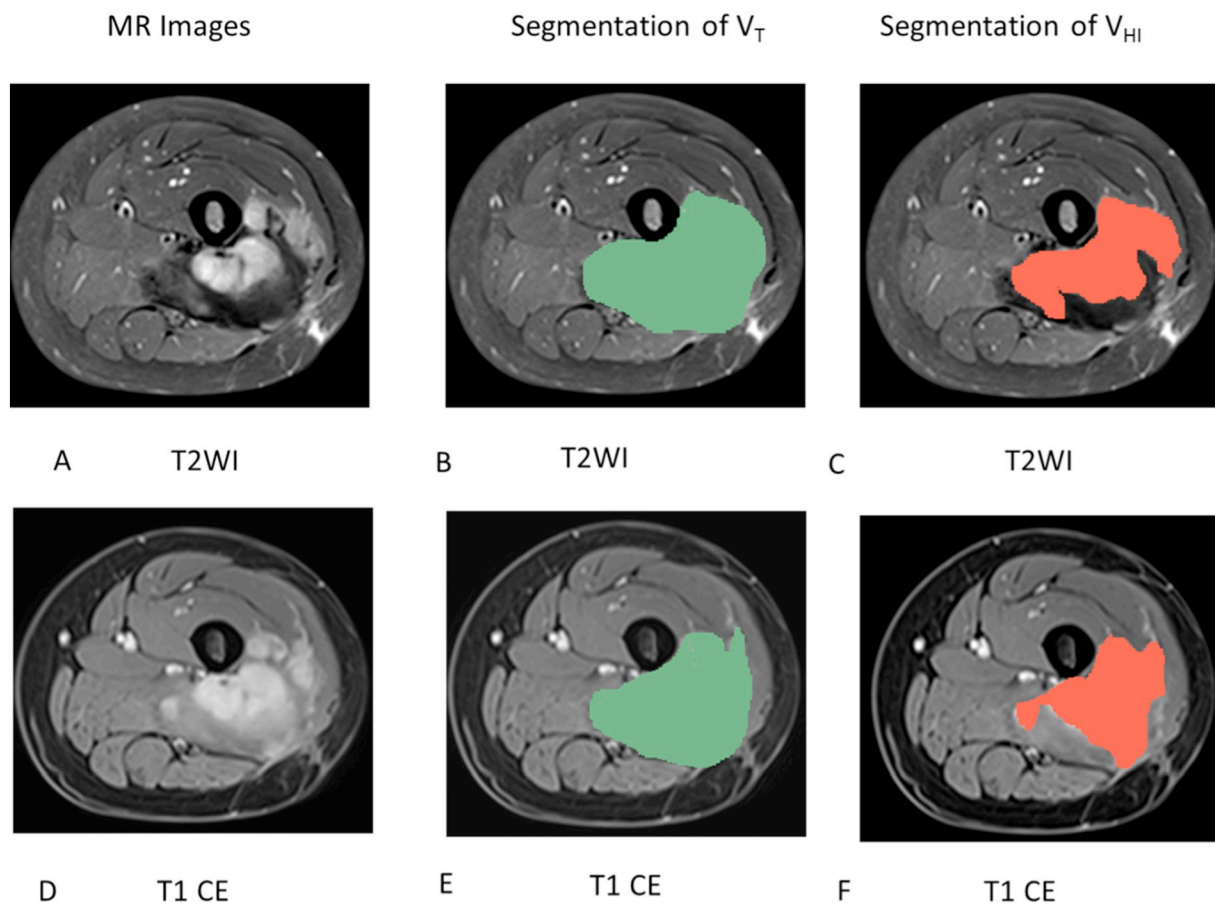


Fig. 1. Schematic diagram of evaluation of MRI hyperintensity. On T2WI, $V_T = 324,868 \text{ mm}^3$ (green zone), $V_{HI} = 266,392 \text{ mm}^3$ (red zone), and $HI = 82.0\%$ (grade = 4). On T1 CE, $V_T = 325,763 \text{ mm}^3$ (green zone), $V_{HI} = 274,618 \text{ mm}^3$ (red zone), $HI = 84.3\%$ (grade = 4). (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

axial T1 CE sequences with fat suppression (Fig. 1). Adjacent normal muscle was used as a reference during the evaluation. Then, the tumour volume (V_T in the following equation) and volume of signal hyperintensity (V_{HI} in the following equation) were calculated from the sum of the areas on the images multiplied by the slice thickness using Slicer 4.8.1 software (Brigham and Women's Hospital, Boston, MA, and Massachusetts Institute of Technology, Cambridge, MA). The hyperintensity percentage of the tumour (HI in the following equation) was determined according to the following formula:

$$HI = \frac{V_{HI}}{V_T} \times 100\%$$

HI was measured by two radiologists and averaged. The signal hyperintensity of each lesion on T2WI and T1 CE images was graded as follows [15,19] (Appendix 2):

- grade 0: $HI = 0\%$, complete signal hypointensity of lesions
- grade 1: $HI = 0\text{--}10\%$, diffuse hypointensity with scant areas of signal hyperintensity
- grade 2: $HI = 11\text{--}50\%$, heterogeneous lesion > 50% hypointense
- grade 3: $HI = 51\text{--}75\%$, heterogeneous lesion < 50% hypointense
- grade 4: $HI = 75\text{--}90\%$, almost complete signal hyperintensity
- grade 5: $HI = 91\text{--}100\%$, complete signal hyperintensity

For the S & S criterion, we used 5%, 10%, 15%, 20% and 30% as the size variation ("X%" in Table 1, labelled as S & S 5%, S & S 10%, S & S 15%, S & S 20% and S & R, respectively). S & R referred size variation of 30% in the PR group and 20% in the PD group. The intensity variation between baseline and follow-up images was recorded, and change in two grades or more was defined as an increase or decrease in MRI signal

intensity. Additionally, if new areas of increased T2 signal intensity and enhancement were detected, the tumour response evaluation was PD.

Before conducting signal grading, the two readers (ZHC and JWY) were familiar with the criterion of MRI signal grade using cases exclusively of this study and then independently conducted signal grading of the baseline MRI using a set of 50 cases, which were randomly selected from a total of 129 cases, to assess the interobserver agreement.

2.5. Reference for tumour response assessment

The evaluation of tumour response using RECIST1.1 at 6th follow-up MR examination at the 36th month since the baseline MR examination was considered as the reference.

2.6. Statistical analysis

Continuous variables were presented as mean $\pm$ standard deviation, while categorical variables were presented as numbers and proportions. The weighted kappa coefficient was calculated for evaluating inter-observer agreement of T2 and T1 grades, 0–0.40, 0.41–0.75, and > 0.75, indicating poor, moderate, and good agreement, respectively. Pearson correlation coefficient or Spearman's rho was calculated, and scatter plots were drawn. Univariate ANOVA or Kruskal-Wallis test was conducted for comparisons among groups. SPSS 22.0 was used for statistical analysis. A P-value < 0.05 was considered statistically significant.

3. Results

3.1. Demographic characteristics

A total of 129 patients underwent 903 MR examinations. The time of six follow-up MR examinations were 6 ± 0.3 months, 12 ± 0.5 months, 18 ± 0.7 months, 24 ± 0.6 months, 30 ± 0.3 months, and 36 ± 0.5 months after baseline MRI. At the 6th follow-up, none of the patients exhibited CR, 61/129 (47.3%) exhibited PR, 52/129 (40.3%) exhibited SD, and 16/129 (12.4%) exhibited PD according to RECIST 1.1. The mean age was 24.7 (range 12–47 years). There were 71 women (55.0%) and 58 men (45.0%). Tumours were located in the upper or lower extremity 44 (34.1%), chest or abdominal wall 12 (9.3%), trunk 67 (51.9%) and intra-abdominal 6 (4.7%). 28 patients had primary tumours, while 101 patients had recurrence tumours. 26 cases had no adjacent structure involvement, while 79.8% cases had various degrees of adjacent structure involvement. 8 patients treated with imatinib; 85 patients treated with chemotherapy (vinorelbine and methotrexate); 28 patients treated with the above two strategies, chemotherapy and imatinib; 8 patients were treated with wait-and-see policy until progressive disease was observed. The two observers obtained good agreement for T2 and T1 grades, with kappa coefficients of 0.793 and 0.782, respectively.

3.2. Correlations between size and signal grade

On baseline MR examination, the maximum diameter was weakly correlated with MRI signal intensity ($r = 0.01$ and -0.057 for T2 grade and T1 CE grade, respectively). T2 grade was weakly correlated with T1 CE grade ($r = 0.28$).

At the 6th follow-up, T2 grade correlated strongly with T1 CE grade ($r = 0.85$). However, MRI signal intensity was weakly correlated with the maximum diameter ($r = 0.36$ and 0.25 for T2 grade and T1 CE grade, respectively).

The variation in lesion signal on T2 grade showed not only a similar tendency to that of T1 CE grade variation but also a correlation with change in maximum diameter. Changes in MRI signal intensity was moderately correlated with changes in maximum diameter ($r = -0.56$ and -0.41 for T2 grade and T1 CE grade, respectively).

3.3. Comparison of size and signal according to tumour response groups

The means and standard deviations of the maximum diameter and MRI signal according to tumour response are presented in [Appendices 3 and 4](#).

During the follow-up period, changes in maximum diameter and T2 grade among the three tumour response categories were significantly different (all $p < 0.01$). At the 1st and 2nd follow-up, changes in MRI intensity of T1 CE among the three tumour response categories were not statistically different ($p = 0.87$ and 0.15 , respectively); however, a significant difference was noted from the 3rd to 6th follow-up MRI (all $p < 0.01$).

3.4. Accuracy of the S & S for tumour response evaluation at each follow-up

The comparison between RECIST1.1 and S & S criterion among the three tumour response categories is presented in [Fig. 2](#). It was found that the S & S criterion could predict tumour response earlier than RECIST1.1, and this result was more obvious in the PR and PD groups. More than 95% of patients in the PR group were predicted by S & S criterion at the 2nd follow-up (58/61, 58/61, and 57/61 for S & S 10%,

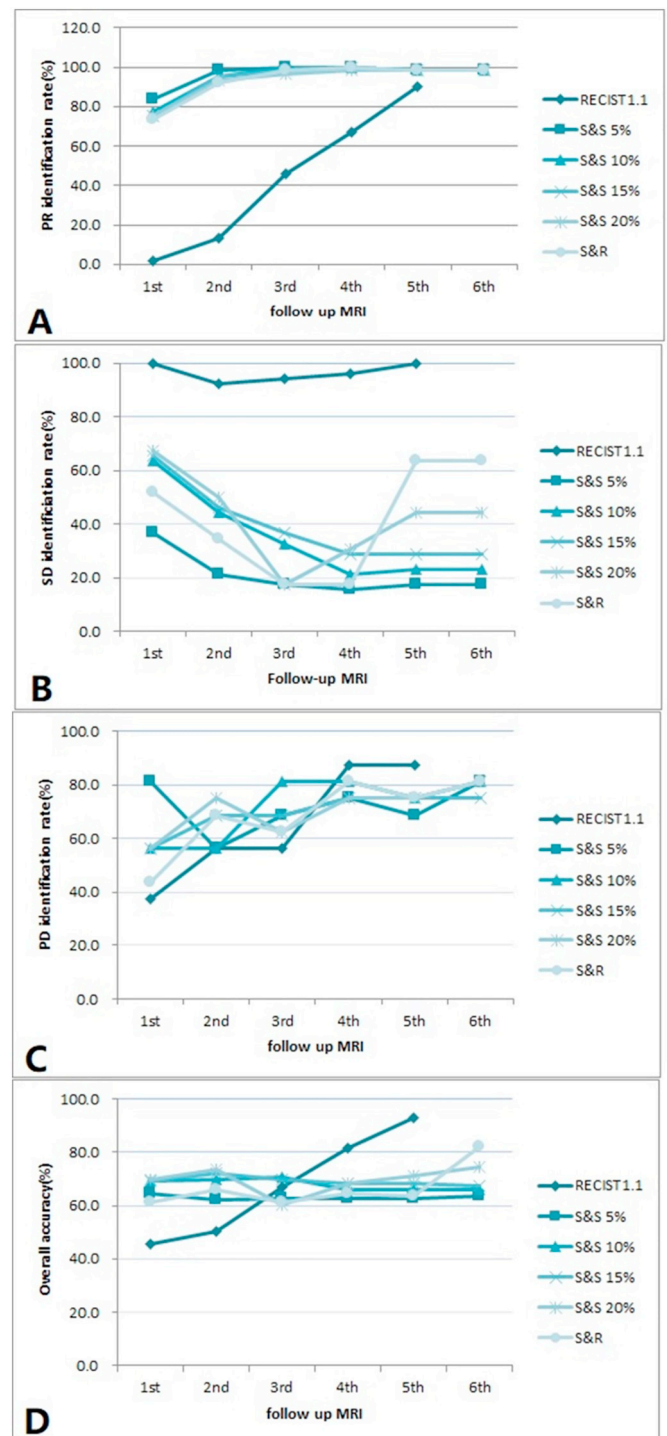


Fig. 2. Comparison between the RECIST1.1 and Signal and Size (S&S) criterion among three tumour response categories, PR (A), SD (B), and PD (C), and overall accuracy (D) are presented. S&S 5%, S&S 10%, S&S 15% and S&S 20% represented signal and 5%, 10%, 15% and 20% size variation; S&R represented signal and 30% size variation for PR, 20% size variation for PD. The labels 1st, 2nd, 3rd, 4th, 5th, and 6th refer to the 6th, 12th, 18th, 24th, 30th, and 36th month of follow-up, respectively. Tumour response assessment at the 6th follow-up MR examination at the 36th month since baseline MR examination according to RECIST1.1 was used as the reference.

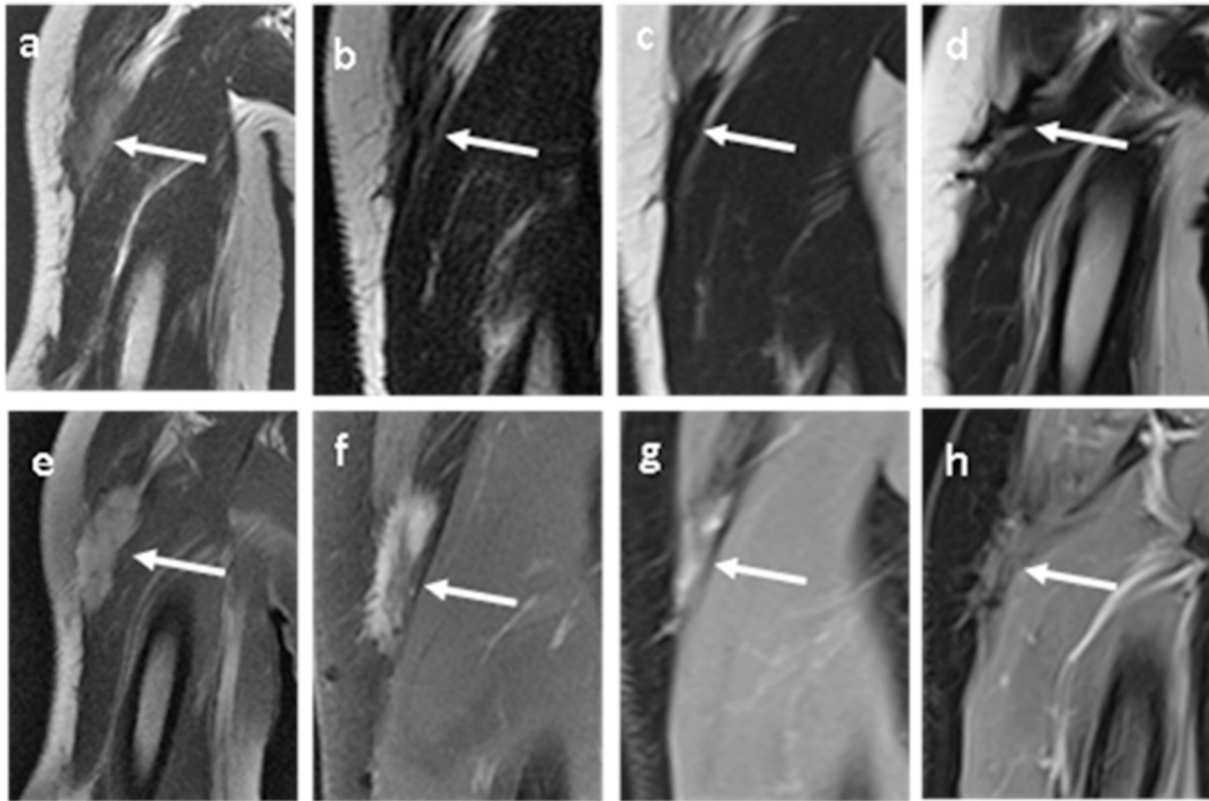


Fig. 3. MRI images in a 38-year-old female patient with DF of the right deltoid muscle without any treatment. a–d. Coronal T2-weighted imaging. e–h. Coronal contrast-enhanced (CE) T1-weighted imaging.

Baseline (a and e): diameter, 41 mm; the lesion has almost completely signal hyperintensity relative to adjacent muscle on T2-weighted imaging (HI = 80%, grade 4) and complete signal hyperintensity on CE (HI = 92.3%, grade 5).

1st follow-up (b and f): diameter, 34 mm, there was a 17.1% decrease in tumour size; note the marked reduction in MRI signal intensity of the lesion, which is > 50% hypointense on T2-weighted imaging (HI = 29.6%, grade 2) and < 50% hypointense on CE (HI = 63.5%, grade 3). The tumour response assessment is SD according to RECIST1.1 and PR according to the S & S criterion.

4th follow-up (c and g): diameter, 22 mm, there was a 46.3% decrease in tumour size; the lesion had diffuse hypointensity with scant areas of signal hyperintensity on T2-weighted imaging (HI = 4.6%, grade 1) and > 50% hypointense on CE (HI = 21.7%, grade 2). The tumour response assessment is PR according to RECIST1.1 and the S & S criterion.

6th follow-up (d and h): diameter, 17 mm, there was a 58.5% decrease in tumour size; the lesion had complete signal hypointensity on both T2-weighted imaging and CE (HI = 0%, grade 0). The tumour response assessment is PR according to RECIST1.1 and the S & S criterion.

S & S 15%, and S & S 20%, respectively), while only 13.1% were predicted by RECIST1.1 (8/61) (Fig. 3). Moreover, 81.2% of patients in the PD group were predicted by S & S criterion at the 3rd follow-up (13/16 for S & S 10%), while only 56.3% were predicted by RECIST1.1 (9/16) (Fig. 4). However, S & S criterion was less effective than RECIST1.1 in the SD group. More than 65% of patients in the SD group were predicted by S & S criterion at the 1st follow-up (33/52, 34/52, and 35/52 for S & S 10%, S & S 15%, and S & S 20%, respectively), while 100% were predicted by RECIST1.1 (52/52). At the 6th follow-up, < 50% of patients in the SD group were predicted by S & S criterion (12/52, 15/52, and 23/52 for S & S 10%, S & S 15%, and S & S 20%, respectively). Therefore, the overall accuracy of the S & S criterion was always < 75%.

3.5. Analysis of false positive rate of the S & S criterion

The tumour response assessment by S & S criterion had an unsatisfactory positive predictive value. In the S & S criterion, the false positive rates for PR were approximately 30–40% at six follow-up time

points; the false positive rates for PD were approximately 20–30%. Thus, a significant proportion of patients in the SD group were mistaken for PR or PD by S & S criterion.

4. Discussion

This study was designed to investigate the impact of treatment on tumour growth and MRI signal variation and the validity of a new criterion of tumour response assessment in patients with DF undergoing non-surgical therapies. Compared with previous studies focusing on using MRI signal to assess the tumour response of DF, this study which combined MR signal and size for assessing tumour response of DF, has its advantage in relatively large number of cases and long follow-up. In 2016, Braschi-Amirfarzan et al. proposed the combination of MRI signals and tumour size to assess the tumour response of DF [1]; however, there is no consensus on how to combine MRI signal intensity and tumour size. We refined Braschi-Amirfarzan's criterion, quantified signal and tumour size changes in target lesions, and applied different size variations to tumour response assessment.

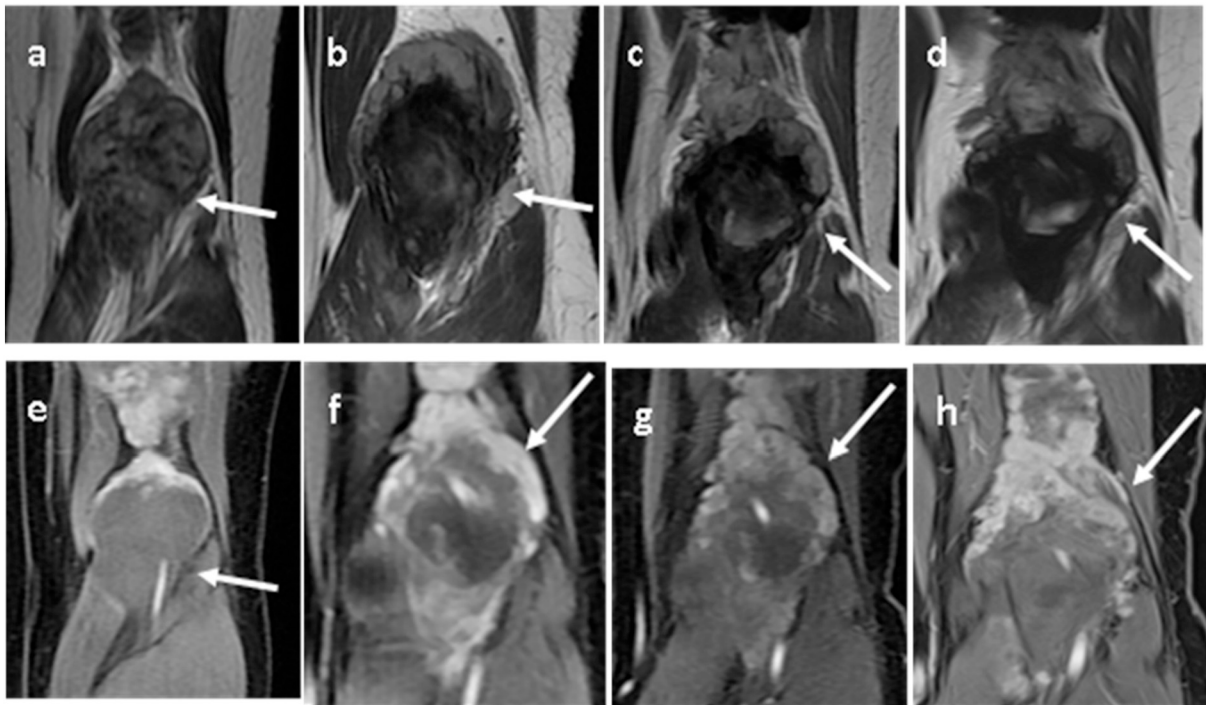


Fig. 4. MRI images in a 22-year-old female patient with DF of the left popliteal fossa before and after chemotherapy. a–d. Coronal T2-weighted imaging. e–h. Coronal contrast-enhanced (CE) T1-weighted imaging.

Baseline (a and e): diameter, 100 mm; the lesion had almost complete signal hyperintensity on T2-weighted imaging (HI = 83.8%, grade 4) and diffuse hypointensity with scant areas of signal hyperintensity on CE (HI = 6.7%, grade 1).

1st follow-up (b and f): diameter, 106 mm, there was a 0.6% increase in tumour size. The new enhancing areas (arrowhead) along the superior aspect of the tumour (grade 5), indicating an aggressive component of the tumour. The tumour response assessment is SD according to RECIST1.1 and PD according to the S & S criterion.

3rd follow-up (c and g): diameter, 115 mm, there was a 15% increase in tumour size; new areas of increased enhancement are still in the lesion (grade 5). The tumour response assessment is SD according to RECIST1.1 and PD according to the S & S criterion.

6th follow-up (d and h): diameter, 135 mm, there was a 35% increase in tumour size and an increase in lesion size. The tumour response assessment is PD according to RECIST1.1 and the S & S criterion.

This study used a hierarchical approach to evaluate MRI signal intensity, which was supported by the literatures [15,19]. Observers in our study calculated the hyperintensity percentage of tumours by manually drawing on each slice of stacked MR images and provided a grade according to the percentage of hyperintensity. Although the evaluation was subjective, the consistency between the observers was good. Hence, this grading system was demonstrated to be applicable to the tumour response assessment of DF.

Although DF lacks metastatic potential, its unpredictable behaviour, propensity for progressive infiltration, and local invasion make treatment challenging [1,6,14,23]. Previous studies have shown that three tumour response categories (PR, SD, and PD) demonstrated different degrees of symptom relief and genetic testing [24,25]. Early detection of patients with PR indicates that the treatment is effective, which may contribute to the development of new treatments [24]. If disease progression is observed, patients with close observations require clinical intervention, and patients with systemic therapy require drug replacement [16,26,27]. These results indicate that early detection of variance in patients' condition contributes to clinical decision-making.

Our findings demonstrated that changes in MRI signal intensity were moderately correlated with changes in maximum diameter ($r = -0.56$ and -0.41 for T2 grade and T1 CE grade, respectively). This result was similar to that of Pooja's study [13], indicating that MRI signal intensity may be applied to tumour response assessment.

According to the different therapeutic effects, the change in MRI signal intensity of the target lesion begins to show a different tendency.

For the PD group, there was a slight change in the MRI signal between baseline and follow-up MR examinations. This result indicates that the proportion of histological components inside the tumour in the growth phase was not significantly different from the baseline, and tumour progression only showed an increase in size. For the SD and PR groups, there was a significant change in MRI signal intensity between baseline and follow-up MR examinations. This demonstrates that there is a decrease in vascularity, spindle cells, cellular stroma, and myxoid matrix and an increase in fibrous tissue and collagen bands, which results in decreased signal intensity on MR images [1,5]. This reflects the fact that tumour responses to treatment of DF are accompanied by a decrease in not only tumour size but also MRI signals. The presented findings are in accordance with those of previous literature reports, which demonstrate that a decrease in T2 signal intensity indicates a positive response to systemic therapy [1,13,19].

We found that the signal and size criterion can predict tumour response earlier than RECIST1.1 in the PR and PD groups. Early detection of patients with PR indicates that the treatment is effective and the patient's condition is stable, thereby increasing patients and physicians' satisfaction and confidence with treatment. Early detection of patients with PD can help change treatment strategies and prevent aggravation of the disease as soon as possible. According to our findings, we recommend that MRI signal intensity should be added to the tumour response assessment at the early follow-up of non-surgical therapy of DF, and the variation in the maximum diameter of the target lesion should be appropriately reduced (10% or 15%) to facilitate the identification

of patients with regression or progression.

However, there are some problems with the signal and size criterion. We adopted RECIST1.1 as size variation and found that the consistent rate of RECIST1.1 and signal and size criterion was always < 85%. Some lesions that exhibited SD were evaluated as having PR or PD by signal and size criterion. This may be due to the special biological behaviour of DF, which is manifested as a change in the proportion of histological components inside the tumour initially, but the change in tumour size is relatively lagging. The MRI signal intensity was significantly reduced in some lesions exhibiting SD, but the maximum diameter slightly changed.

Our study had a number of limitations. We could not correlate tumour response with overall survival, as in previous studies [28–30], because DF is not a malignant tumour and rarely causes death at a 3-year follow-up. Hence, we selected long-term RECIST1.1 as the reference standard for assessing treatment response, as is currently employed in clinical trials [5]. The limitation of using RECIST1.1 as a reference, mentioned above, may define tumours with biologic burnout but little size change as the SD category. However, there might actually be a PR case as the signal and size criterion is defined (Fig. 3 and Appendix 8) [1,31,32]. Additionally, because of neither guidelines for DF nor recommendation of MRI protocol from the recent European consensus on DF [5], other MR markers, such as quantitative diffusion-weighted or T2-mapping parameters, are excluded in our study. There is no evidence that patients will benefit from these additional MRI sequences. Diffusion-weighted or T2-mapping MRI sequences are also not used in the recent literatures on DF of leading journals [1,16,33,34]. These advanced MRI techniques may be used in the future to provide

additional quantitative parameters for tumour characterisation.

Therefore, this study verified that MRI signals are useful to assess the tumour response of DF, and they could be added to establish a specified signal and size criterion. The signal and size criterion can identify patients with PR and PD earlier than RECIST1.1, which would promote timely adjustment of treatment and be potentially applicable in clinical practice.

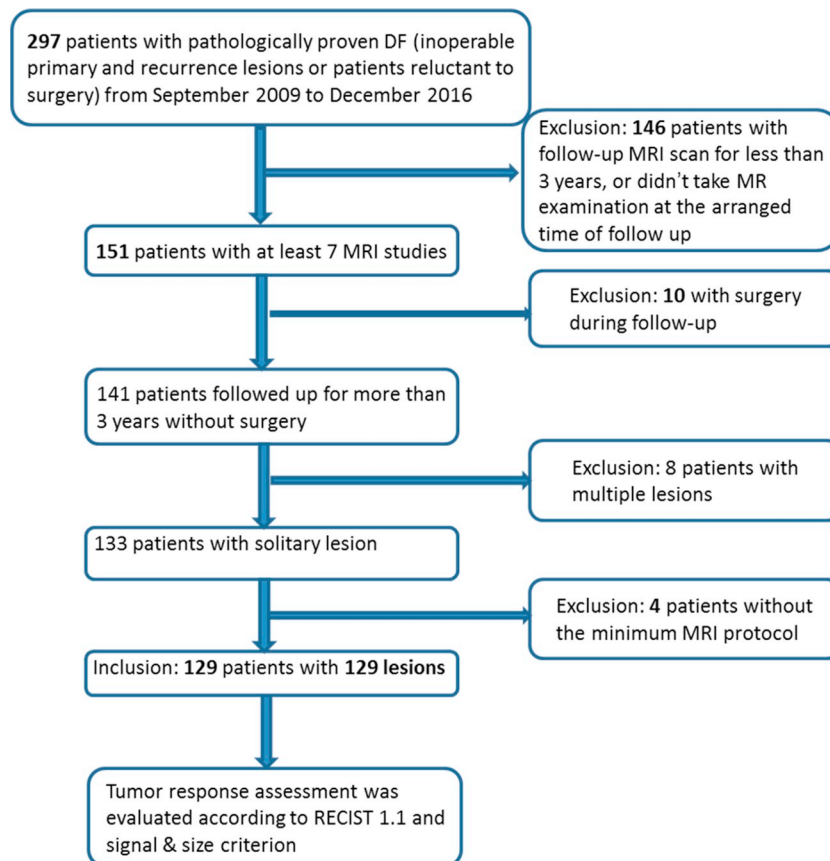
Declaration of competing interest

All authors disclosed no relationship.

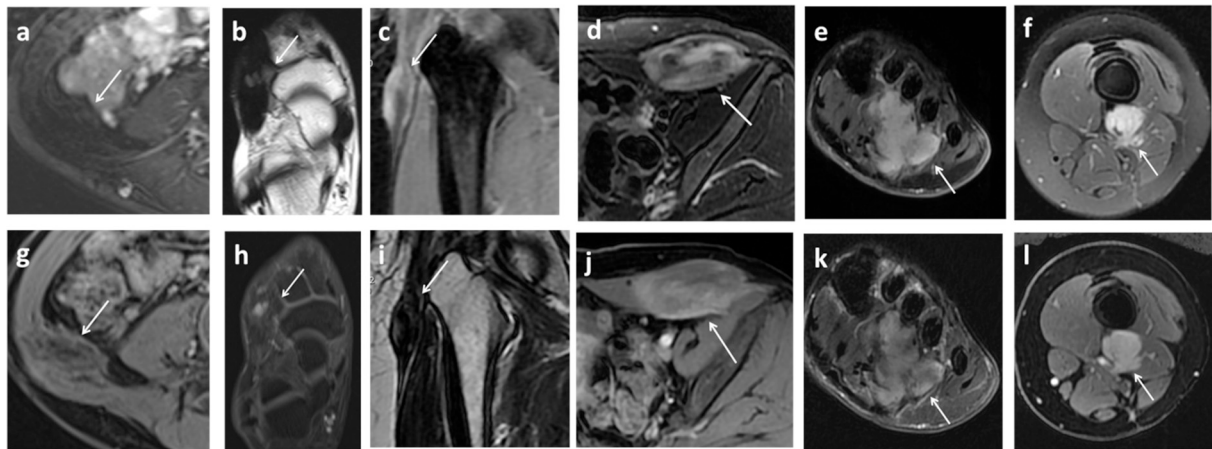
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Study design: H.C.Z, Y.S.S.; provision of study materials or patients: H.C.Z, S.L.; data processing, analysis, and interpretation: H.C.Z, X.T.L, W.Y.J, Y.S.S; manuscript writing and editing: all authors.



Appendix 1. Flowchart of the study methodology and processes.



Appendix 2. Examples of MRI signal intensity grade. a–f: T2-weighted imaging. g–l: contrast-enhanced T1-weighted imaging with fat suppression.

a and g (grade 0): HI = 0%, complete signal hypointensity

b and h (grade 1): HI = 0–10%, diffuse hypointensity with scant areas of signal hyperintensity

c and i (grade 2): HI = 11–50%, heterogeneous lesion > 50% hypointense

d and j (grade 3): HI = 51–75%, heterogeneous lesion < 50% hypointense

e and k (grade 4): HI = 75–90%, almost complete signal hyperintensity

f and l (grade 5): HI = 91–100%, complete signal hyperintensity

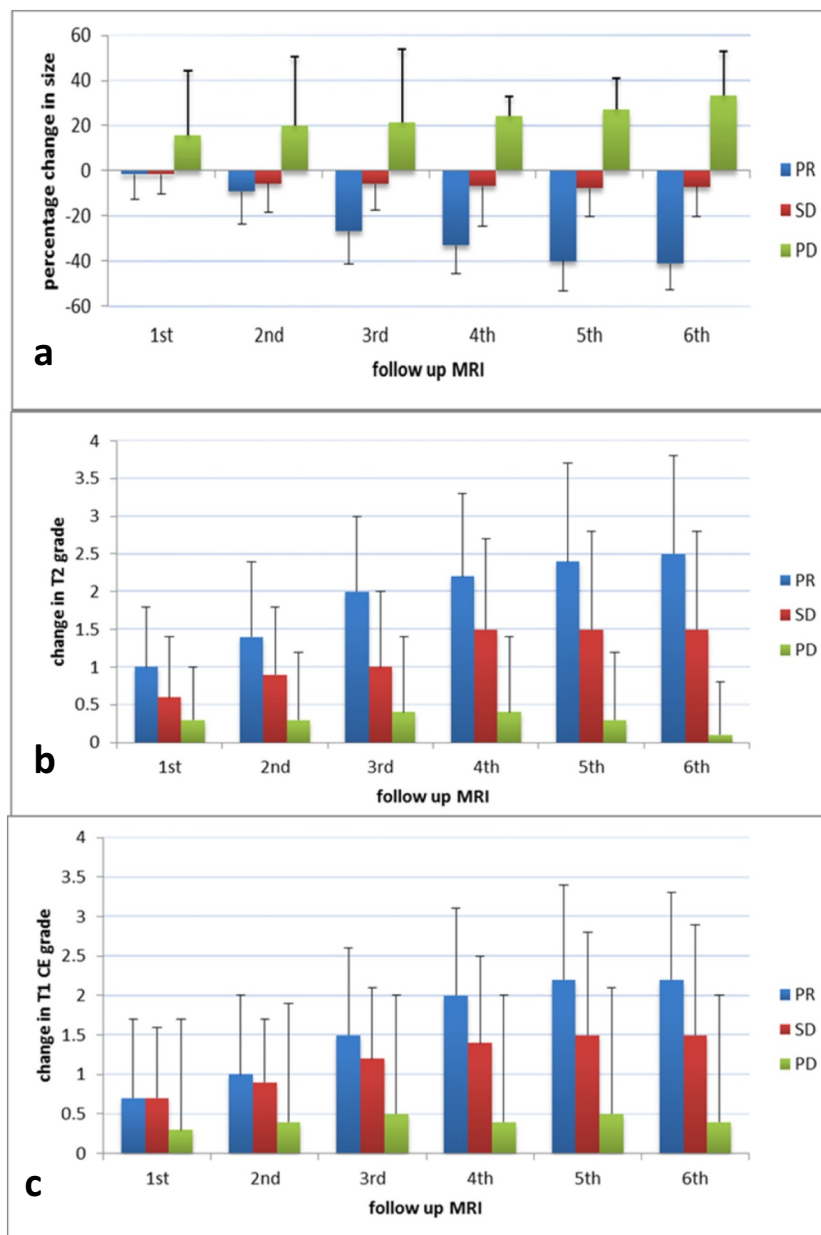
Appendix 3

Variation in the maximum diameter (Dmax) and signal intensity grades of desmoid-type fibromatosis according to tumour response assessment.

Follow-up		Tumour response assessment			p-Value
		PR(N = 61)	SD(N = 52)	PD(N = 16)	
Baseline	Dmax	80.6 ± 40.1	93.9 ± 50.1	77.4 ± 39.3	< 0.01**
	T2 grade	3.8 ± 0.8	3.9 ± 0.4	4.0 ± 0.3	0.21
	T1 CE grade	3.7 ± 1.0	4.0 ± 0.8	4.0 ± 1.1	0.70
1st	Dmax*	−1.6 ± 10.9	−1.5 ± 8.8	15.6 ± 28.8	< 0.01**
	T2*	1.0 ± 0.8	0.6 ± 0.8	0.3 ± 0.7	< 0.01**
	T1*	0.7 ± 1.0	0.7 ± 0.9	0.3 ± 1.4	0.87
2nd	Dmax*	−9.3 ± 14.5	−5.6 ± 12.7	19.8 ± 30.5	< 0.01**
	T2*	1.4 ± 1.0	0.9 ± 0.9	0.3 ± 0.9	< 0.01**
	T1*	1.0 ± 1.0	0.9 ± 0.8	0.4 ± 1.5	0.15
3rd	Dmax*	−26.9 ± 14.2	−5.8 ± 11.6	21.3 ± 32.4	< 0.01**
	T2*	2.0 ± 1.0	1.0 ± 1.0	0.4 ± 1.0	< 0.01**
	T1*	1.5 ± 1.1	1.2 ± 0.9	0.5 ± 1.5	< 0.01**
4th	Dmax*	−33.2 ± 12.3	−6.9 ± 17.8	24.1 ± 8.8	< 0.01**
	T2*	2.2 ± 1.1	1.5 ± 1.2	0.4 ± 1.0	< 0.01**
	T1*	2.0 ± 1.1	1.4 ± 1.1	0.4 ± 1.6	< 0.01**
5th	Dmax*	−40.1 ± 12.9	−7.7 ± 12.5	27.3 ± 13.6	< 0.01**
	T2*	2.4 ± 1.3	1.5 ± 1.3	0.3 ± 0.9	< 0.01**
	T1*	2.2 ± 1.2	1.5 ± 1.3	0.5 ± 1.6	< 0.01**
6th	Dmax*	−41.3 ± 12.4	−7.1 ± 13.1	33.4 ± 39.4	< 0.01**
	T2*	2.5 ± 1.3	1.5 ± 1.3	0.1 ± 0.7	< 0.01**
	T1*	2.2 ± 1.1	1.5 ± 1.4	0.4 ± 1.6	< 0.01**

Dmax* = variation in the maximum diameter and the unit of Dmax* is mm; T2* = variation in T2 grade; T1* = variation in T1 CE grade. The labels 1st, 2nd, 3rd, 4th, 5th, and 6th refer to the 6th, 12th, 18th, 24th, 30th, and 36th month of follow-up, respectively.

** Statistically significant.



Appendix 4. Histogram shows the relative change in the maximum diameter (a), T2 grade (b), and T1 CE grade (c) for lesions stratified by tumour response. During the follow-up period, changes in maximum diameter, T2 grade, and T1 CE grade among the three tumour response categories demonstrated a statistically significant difference ($p < 0.01$) since the 3rd follow-up examination. The labels 1st, 2nd, 3rd, 4th, 5th, 6th refer to the 6th, 12th, 18th, 24th, 30th, and 36th month of follow-up, respectively.

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